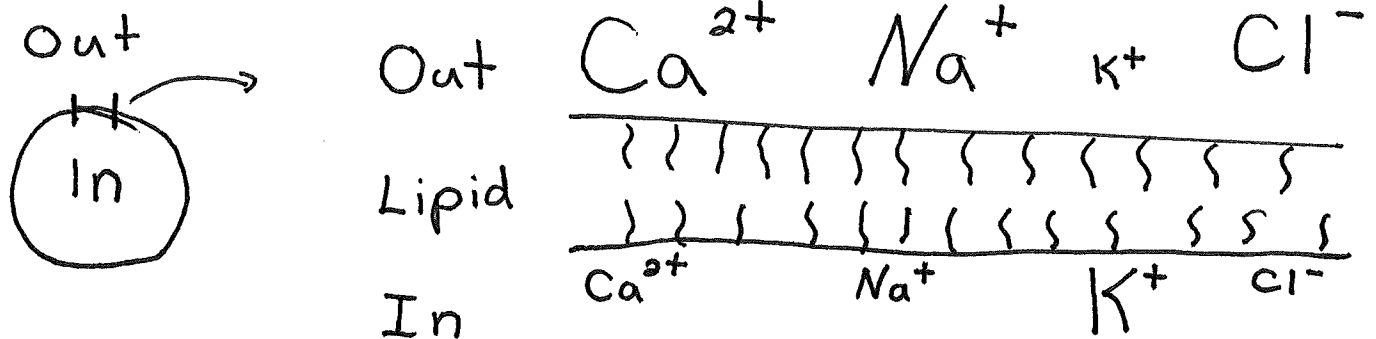
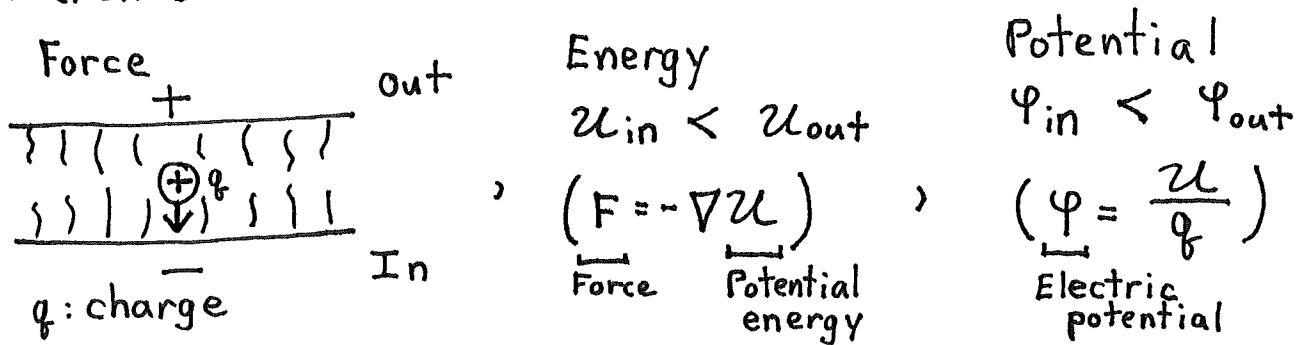


Neurons

Neurons are the fundamental cells of the nervous system. Like all cells, they are surrounded by an impermeable membrane that allows the contents of a neuron to differ from the extracellular space.



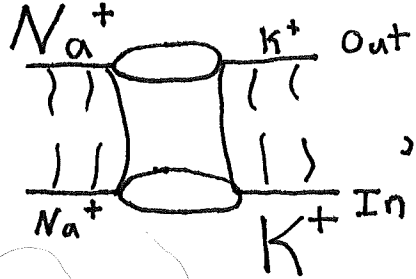
The separation of charges by the membrane leads to an electrical force and potential difference across the membrane.



The membrane voltage is the potential difference across the membrane

$$V_m = \varphi_{in} - \varphi_{out} \approx -60 \text{ mV at rest}$$

It is negative at rest. However, it changes over time because ion-selective holes in the membrane, called ion channels, open and close. The net current that flows depends on electric force across the membrane and the concentration difference of the ion, which is reduced by diffusion.



Na channel:
 Electric: In
 Diffusion: In
 Current: In
 Voltage: Up

K channel:
 Electric: In
 Diffusion: Out
 Current: (Out)
 Voltage: (Down)

Current flows and the voltage changes until an equilibrium is reached

$$P_{\text{state}} \propto e^{-U_{\text{state}}/(kT)} \Rightarrow \frac{P_{\text{in}}}{P_{\text{out}}} = \frac{e^{-q\psi_{\text{in}}/(kT)}}{e^{-q\psi_{\text{out}}/(kT)}} = e^{-qV_m/(kT)}$$

Boltzmann distribution
 T: temperature
 k: Boltzmann constant
 [k = 1.38 · 10⁻²³ joules per Kelvin]

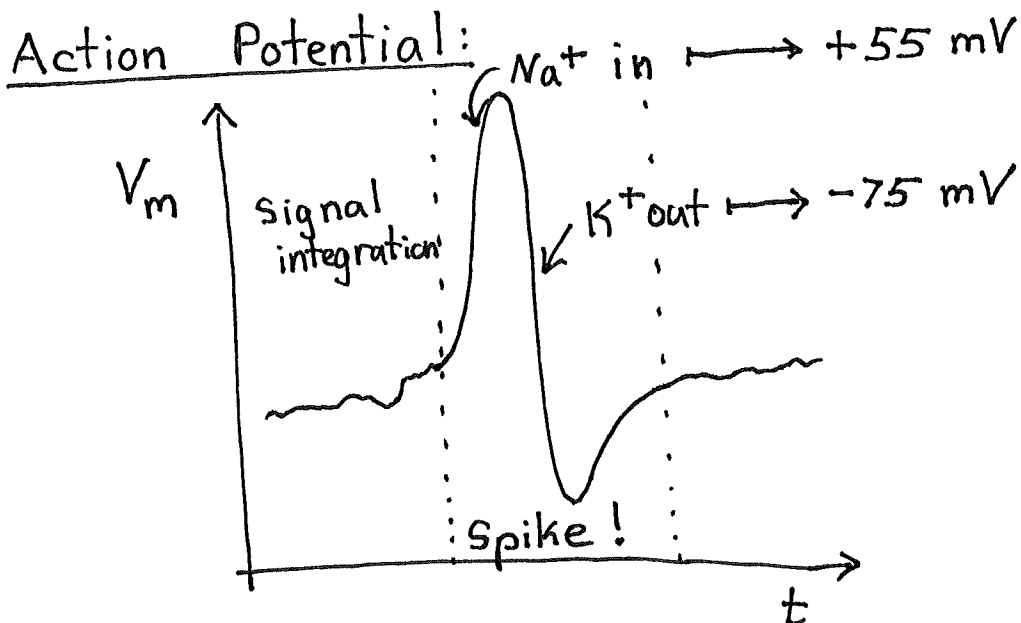
$$\Rightarrow \frac{[C]_{\text{in}}}{[C]_{\text{out}}} = e^{-q_c E_c / (kT)} \quad \left[\begin{array}{l} \text{For ion C} \\ \text{with charge } q_c \end{array} \right]$$

$$\Rightarrow E_c = \frac{kT}{q_c} \ln \left(\frac{[C]_{\text{out}}}{[C]_{\text{in}}} \right)$$

Nernst/Reversal/Equilibrium potential

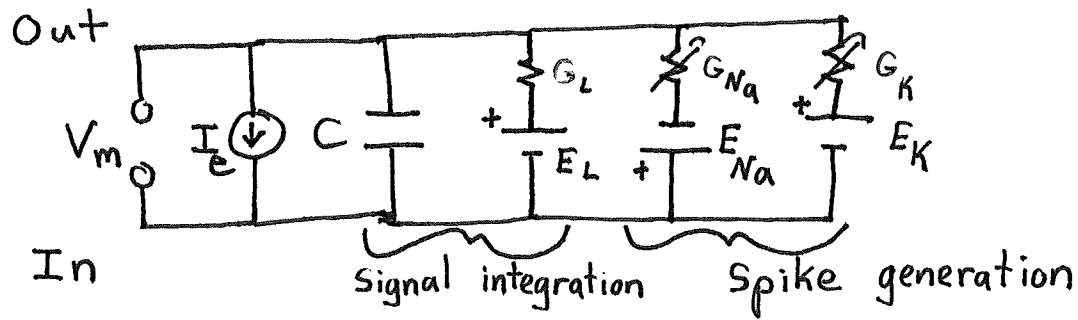
Given physiological temperatures and concentrations:

$$E_{Na} \approx 55 \text{ mV}, \quad E_K \approx -75 \text{ mV}, \quad E_{Ca} \approx 125 \text{ mV}, \quad E_{Cl} \approx -60 \text{ mV}$$



What mechanisms cause an action potential (spike) in the squid giant axon? Answering this question earned Alan Hodgkin and Andrew Huxley the Nobel Prize in 1963.

Electric circuit model: "Hodgkin Huxley"



$$V_m = \varphi_{in} - \varphi_{out}$$

I_e : external current

C : Capacitance

G_L : "Leak" conductance ($E_L = -65 \text{ mV}$)

G_{Na} : Variable sodium conductance

G_K : Variable potassium conductance

Kirchhoff's Current law:

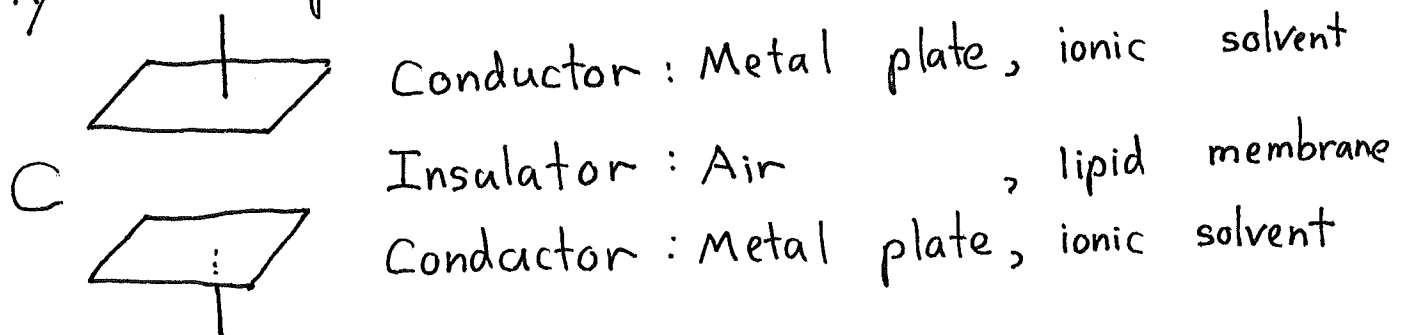
$$\sum_n I_n = 0 \quad (\{I_n\}: \text{Currents flowing in or out of node})$$

Conservation of charge

What if a neuron was just a capacitor?

$$I_e - I_c = 0$$

Why a capacitor?



Capacitors separate charge:

$$Q = CV$$

Capacitive current:

$$I_c = \frac{dQ}{dt} = C \frac{dV}{dt}$$

"Capacitor model" of neuron:

$$C \frac{dV_m}{dt} = I_e$$

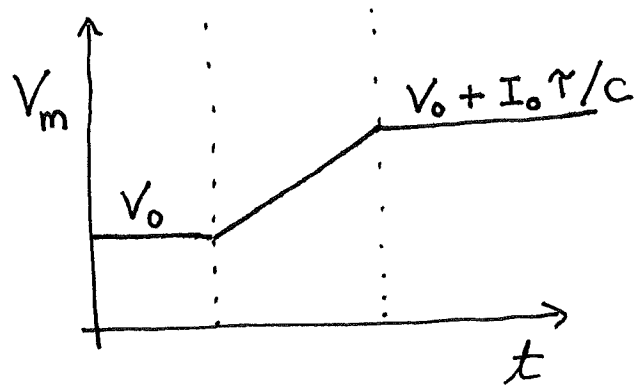
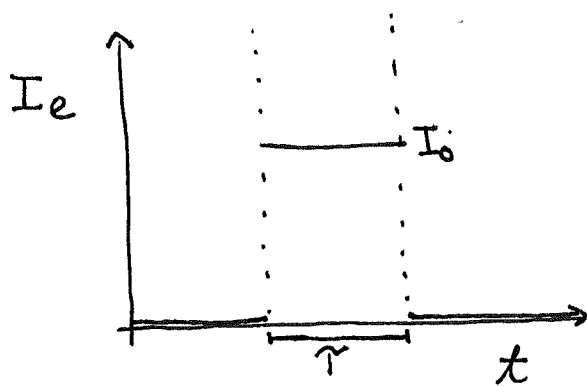
Current integration in "capacitor model" of neuron.

$$C \frac{dV_m}{dt} = I_e$$

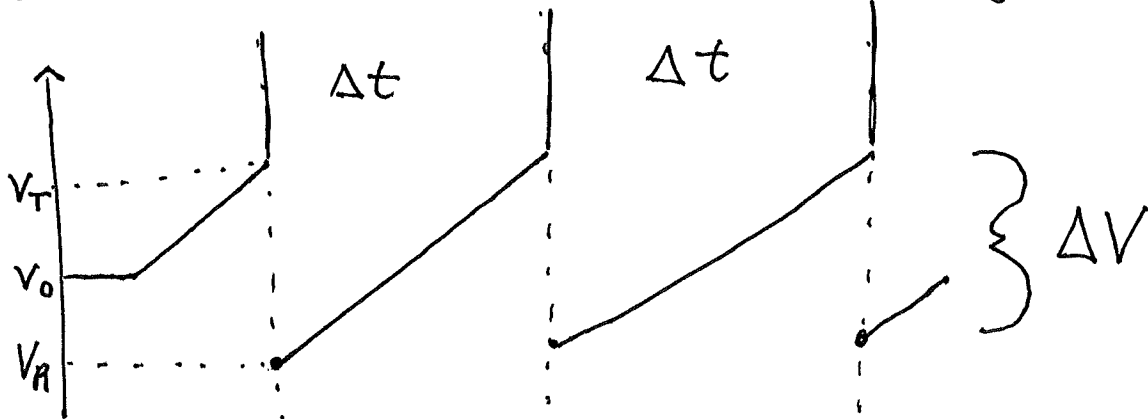
$$V_m(t) = V_m(0) + \frac{1}{C} \int_0^t dt' I_e(t')$$

$$\Delta V_m = \frac{\Delta Q}{C}$$

Response to a current step:



A simple model of neuronal firing: Integrate & Fire



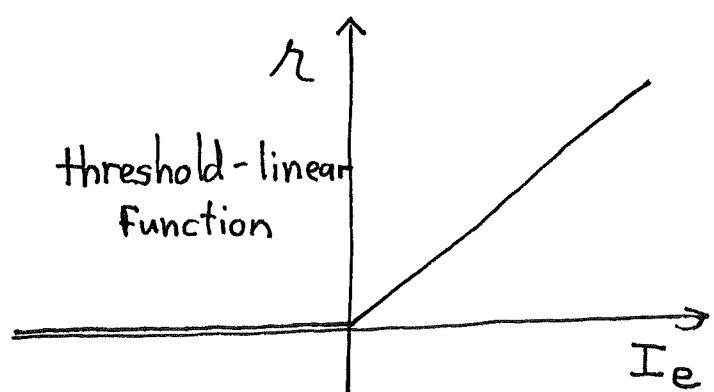
Firing rate:

$$h = \frac{1}{\Delta t}$$

$$\Delta V = V_T - V_R = \frac{I_e \Delta t}{C}$$

$$h = \frac{I_e}{C(V_T - V_R)}$$

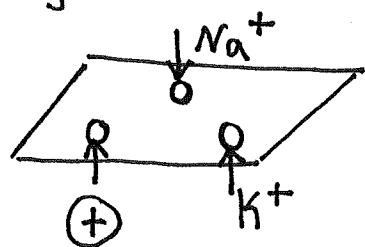
F-I curve



Resistance $\Rightarrow$ Leaky integrate & fire

$$I_e - I_c - I_L = 0$$

Why isn't resistance infinite? Holes let charge through. This is termed leak.

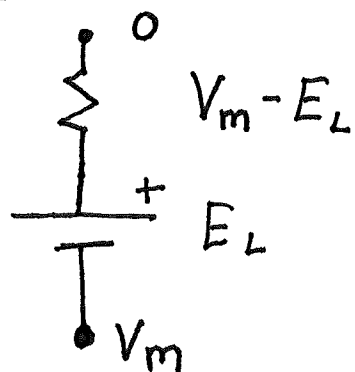


Different holes
Different charged ions $\Rightarrow E_L \neq 0$
(Battery)

Ohm's law:

$$I_L = \frac{V}{R_L}$$

Resistor & battery



$$V_m = I_L R_L + E_L$$

$$I_L = \frac{V_m - E_L}{R_L} = G_L (V_m - E_L)$$

"Leaky integrator model" of neuron:

$$C \frac{dV_m}{dt} = I_e - G_L (V_m - E_L)$$